



Proof-of-Control: An Open Standard for Runtime Verifiability and Cryptographic Oversight in Autonomous AI Execution

— Prove, rather than promise, what your agents did.

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ABSTRACT

Autonomous AI agents plan multi-step workflows, invoke external APIs, modify application state, and delegate sub-tasks with minimal real-time human supervision—exposing a systemic vulnerability we call the *Verifiability Gap*: the structural impossibility of confirming that an agent operated within its authorized policy boundaries and that its controls remained active and unmanipulated. Legacy mechanisms (system prompts, static access control, embedded guardrails, post-hoc log auditing) fail against non-deterministic, path-dependent execution at machine speed. We introduce Proof-of-Control (PoC), an open standard for runtime verifiability comprising: (i) six *domains of verification* enumerating 111 auditable requirements; (ii) four *Verifiability Tiers* grading evidence by who must be trusted to believe it, with a *binary threshold* separating authenticated documentation from mechanism-generated evidence; (iii) a reference architecture coupling lifecycle interception hooks with hardware-isolated policy evaluation under IETF RATS (RFC 9334), emitting Entity Attestation Tokens over a signed Merkle execution chain; and (iv) a reproducible coverage mapping showing that eight leading frameworks (NIST AI RMF, ISO/IEC 42001, OWASP AISVS, EU AI Act, SOC 2, CSA AARM, NIST SP 800-207, MITRE ATLAS) address 27–64% of PoC requirements—almost entirely as partial matches that require the control but not operator-independent evidence of it. We state a formal policy model, a zero-trust threat model, a sub-15 ms latency budget, and an integration roadmap across ISO/IEC JTC 1/SC 42, IETF RATS, and NIST AI 600-1.

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1 Introduction: The Verifiability Gap

The rapid evolution from passive natural-language generation to autonomous agentic execution represents a foundational shift in computational infrastructure [1, 2]. Autonomous agents now plan multi-step operational workflows, query complex databases, invoke external APIs, modify application state, and delegate sub-tasks to secondary agents with minimal real-time human supervision. While this autonomy unlocks substantial productivity, it exposes a systemic architectural vulnerability: the *Verifiability Gap*—the absence of independent evidence of what an AI system did [3].

The Verifiability Gap describes the structural impossibility of confirming—either during execution or retrospectively—that an autonomous agent operated within its authorized policy boundaries, that mandatory security controls remained active and uncompromised, and that the agent’s state transitions were free from malicious manipulation or reasoning failure. Because agents execute complex, non-deterministic actions at machine speed, human oversight and traditional log-auditing workflows cannot maintain effective operational control [4]. Existing governance implementations rely almost exclusively on operator self-attestation, static permissions, or unanchored application logs, none of which provide open, cryptographically verifiable proof of policy compliance [5].

Path dependency. Unlike legacy software executing deterministic decision trees, agents operate within LLM reasoning loops that yield non-deterministic, path-dependent trajectories [5]. Given a single objective, an agent dynamically determines its intermediate steps, selects tools, interprets unstructured outputs, and modifies its path from real-time observations. Compliance therefore cannot be validated prior to deployment, and evaluating an individual tool call in isolation fails to mitigate path-dependent risk: an agent reading confidential records early and issuing an outbound network request later creates a cumulative exfiltration risk that single-step inspection cannot detect. Recent empirical work quantifies this: under skill composition, attack success on capability-flow paths rises from near zero in isolation to 33.6%, trust-transfer attacks succeed on more than 96.5% of attempts across four of five model backends, and contextual contamination raises risky approvals by 71.8% [6]. Mid-trajectory monitoring competence, meanwhile, correlates strongly with structural trace parsing ($\rho = 0.79$ with JSON validation accuracy) and near zero with natural-language safety benchmarks [7].

Contributions. This paper (1) defines the Proof-of-Control standard—six domains of verification, four evidence properties, four Verifiability Tiers with a binary threshold, four requirement levels aligned to the tiers, and three conformance stages—as a catalogue of 111 auditable requirements; (2) specifies a reference architecture integrating lifecycle interception with hardware-backed attestation under IETF RATS [8], Entity Attestation Tokens [9], and signed Merkle execution chains [10]; (3) formalizes the policy model and threat model; (4) reports a reproducible, requirement-level coverage mapping against eight regulatory and security frameworks using the coding methodology of HAARF [11]; and (5) lays out an integration roadmap across ISO/IEC JTC 1/SC 42 [12, 13], the IETF RATS working group [14], the Agent Control Specification [15, 16], and NIST AI 600-1 [17].

2 Why Legacy Governance Fails

Current AI risk management relies on three primary control mechanisms, all of which fail to resolve the Verifiability Gap in agentic environments.

System prompts and in-context instructions. Prompt-level instructions inhabit the same logical stream as untrusted user inputs, retrieved documents, and tool outputs; this co-mingling renders them vulnerable to direct and indirect prompt injection. Prompting provides a probabilistic steering mechanism rather than a deterministic boundary: if a model deviates from its system prompt, no execution barrier prevents the action [5].

Role-based access control and static identity. RBAC restricts an agent from calling unapproved endpoints but operates without contextual awareness. An agent holding valid credentials for both internal database reads and external messaging retains the technical capability to exfiltrate data; static IAM cannot evaluate the context, sequence, or cumulative risk of individually authorized permissions [18].

Application-embedded guardrails. Guardrails compiled into the agent’s application code run under the agent’s own execution authority. For agents with code-generation or system-command tools, internal guardrails provide weak isolation—a compromised process can bypass its own application-level checks—and they generate local, unauthenticated logs that cannot be audited independently of the host operator [19].

Collapse of the post-hoc audit window. The operational window for an autonomous exploit has collapsed toward zero: indirect prompt injections embedded in public web pages or internal repositories can hijack a reasoning loop instantly [2]. Retrospective SIEM auditing is insufficient because application logs lack cryptographic immutability; an operator—or an adversary with persistence—can modify, append, or delete entries post-execution. Verification must therefore shift from reactive log auditing to inline, machine-speed cryptographic attestation anchored to hardware roots of trust [20, 21]. Telemetry-first governance research reaches the same conclusion from the enterprise side: continuous, machine-collected control assertions displace questionnaire-based self-attestation [22], and financial-disclosure analysis shows that narrative disclosure alone does not constitute proof that controls operate [23].

3 The Proof-of-Control Standard

Proof-of-Control is *open verification* that the controls governing an agent system are implemented and hold, graded by how independently that can be verified. For each control it claims, across six domains, a conformant implementation MUST produce evidence—generated by the enforcing mechanism at execution time—that the control held; place that evidence on the Verifiability Tiers; and disclose its residual trust assumptions. A system has Proof-of-Control when, and only when, its evidence reaches Tier 3 or Tier 4 [3]. Requirements language follows RFC 2119 [24].

3.1 Six domains of verification

The standard enumerates 111 requirements across six domains, each stated as a testable *verify-that* obligation with auditor evidence guidance:

1. **Provenance** (C1): which model ran; lineage, artifact, and supply-chain origin, bound by hash-linked custody chains and signed manifests [25, 26]. Cross-organizational deployments realize this pattern with identity-anchored service passports [27].

2. **Privacy** (C2): what data was read and written, evidenced without re-leaking the data being protected (derived evidence, zero-knowledge policy proofs, commitments) [28].
3. **Portability** (C3): boundary crossings—organizational, jurisdictional, compute—with signed linking records so the evidence chain survives migrations.
4. **Authorization** (C4): authority granted, decisions within or against it, delegation validity; *path-aware* evaluation so a composed sequence of individually authorized calls cannot exceed the authority of its steps [6].
5. **Identity** (C5): which agent and which principal ran, bound by short-lived delegation tokens, W3C DIDs and Verifiable Credentials [29, 30], extending to N -party verifiable trust constructs for multi-agent chains [31].
6. **Security** (C6): integrity of the execution environment, controls held, tools invoked, key lifecycle in hardware-backed custody.

Four cross-cutting chapters govern evidence generation and properties (binary, contemporaneous, tamper-evident, transparent), tier grading, system-surface location on the seven MAESTRO layers [32], and conformance with standardized trust-assumption disclosure.

3.2 Verifiability Tiers and the binary threshold

Evidence is graded by *who must be trusted to believe it* (Table 1). Cryptography alone does not raise the tier; removing the trusted party does. An operator-signed log is cryptographic and still Tier 2. A vendor-rooted TEE attestation is Tier 2 unless composed with independent anchoring (e.g., a public transparency log). The *binary threshold* falls between Tier 2 and Tier 3: below it, authenticated documentation; above it, mechanism-generated evidence. The threshold makes the category procurable—“does your AI have Proof-of-Control?” is a yes-or-no question, the pattern used by every category-defining certification (FIPS 140, Common Criteria, PCI DSS, CSA STAR).

Tier	Name	Who you must trust	Proof-of-Control?
1	Assertion	The operator	No
2	Attestation	A third party or root-keeper	No
3	Independently Verifiable	The cryptographic mechanism	Yes
4	Self-Enforcing	No one—cannot run if integrity breaks	Yes

Table 1: The four Verifiability Tiers. The binary threshold falls between Tiers 2 and 3. Requirement levels 1–4 align one-to-one with the tiers; meeting all Level 1–3 requirements in the claimed domains is the minimum for a Proof-of-Control claim, and Level 4 corresponds to self-enforcing, continuously monitored operation.

Conformance is graded on a separate axis by three named stages—Self-Declared, Third-Party Assessed, Continuously Monitored—each requiring a standardized conformance statement with declared system boundary, in-scope action classes, per-claim tiers, mechanisms, and trust-assumption disclosure. The declared inventory **MUST** be reconciled against automated discovery from observability streams so undeclared (“shadow”) agents surface as findings [22].

4 Reference Architecture

The reference architecture integrates runtime lifecycle interception with hardware-backed attestation under the IETF RATS architecture (RFC 9334) [8, 33]:

```

[Agent Host / Runtime]
|
+--> Interception Point (ACS Lifecycle Hook)
|
+--> Canonical Input Builder
    (State Snapshot: Path, Identity, Proposed Action)
    |
    +--> Hardware TEE Enclave (Attesting Environment)
        |-- Evaluates Portable Rego Policy Bundle
        |-- Renders Verdict: ALLOW / DENY / MODIFY / ESCALATE
        |-- Updates Signed Merkle Execution Chain
        |-- Emits Signed Entity Attestation Token (EAT / CWT)
        |
        +--> [Relying Party / Verifier Checks Proof]

```

4.1 Lifecycle interception hooks

Proof-of-Control mandates standardized middleware hooks across the agent execution loop, leveraging the open Agent Control Specification (ACS) [15, 16]: *task initialization*, *context assembly* (RAG/memory injection), *plan generation*, *pre-call tool invocation*, *post-call tool result*, *memory write*, *sub-agent delegation* (with constraint inheritance), and *task completion*. At each point the host constructs a canonical state snapshot—agent identity, execution history, proposed action, environmental context—submitted to an isolated policy engine that returns a binding decision: ALLOW, DENY, MODIFY (inline sanitization), or ESCALATE (human-in-the-loop authorization). The gateway does not forward an action until the corresponding evidence record is durably written, and in-scope actions fail closed when evidence cannot be generated at the claimed tier.

4.2 RATS role mapping

The runtime maps onto RATS roles [8]: the *Attester* is the platform housing the agent runtime, hooks, and policy engine; the *Target Environment* is the operational layer containing the runtime, model context, and proposed action; the *Attesting Environment* is a cryptographically isolated TEE (Intel TDX, AMD SEV-SNP, NVIDIA Confidential Computing) [20, 21] that measures the policy-engine binaries, verifies the policy-bundle signature, evaluates the snapshot, and signs the proof; *Evidence* is an Entity Attestation Token (CWT/JWT) carrying measurements, claims, step index, verdict, and platform signatures [9, 34]; a *Verifier* compares Evidence against reference values to produce an Attestation Result; and *Relying Parties* (API gateways, downstream services) require valid results before executing tool requests. Sub-agent delegation chains are governed by composite multi-verifier topologies [35].

4.3 Merkle-chain execution history

Each intercepted step t generates a leaf

$$L_t = H(L_{t-1} \parallel H(\sigma_t^{\text{snap}}) \parallel d_t), \quad L_0 = H(\text{seed}_{\text{init}}), \quad (1)$$

where σ_t^{snap} is the canonical snapshot, d_t the verdict, and H a collision-resistant hash. The Attesting Environment signs each leaf with an enclave-bound key; per-source monotonic sequence indices make *omission* as detectable as alteration. The accumulated root is periodically anchored to a public ledger or distributed trust root (asynchronously, off the hot path), preventing selective editing, reordering, or suppression of history [10].

4.4 Canonical evidence schema

Evidence is an EAT profile for autonomous agents:

```
{
  "iss": "https://verifier.trusted-attestation.org",
  "iat": 1773532800,
  "nonce": "0x8f93e2a110c49b",
  "eat_profile": "https://standards.org/poc/v1",
  "poc_claims": {
    "agent_id": "did:web:enterprise.com:agents:finance-agent-04",
    "initiating_user": "user:auth0|84f12a9e",
    "agbom_digest": "sha256:7f8a...5f6a",
    "interception_point": "PRE_CALL_TOOL_INVOCATION",
    "step_index": 3,
    "merkle_root": "0x2c3d...2c3d",
    "policy_bundle_hash": "sha256:e3b0...b855",
    "target_resource": "api.bank.com/v1/payments",
    "canonical_snapshot_hash": "sha256:a4b3...a1b2",
    "verdict": "ALLOW"
  },
  "submods": {
    "hardware_attestation": {
      "platform": "INTEL_TDX",
      "mr_config": "0x1a2b3c4d5e6f...",
      "attestation_signature": "0x30440220..."
    }
  }
}
```

Attribute	System prompts	Static RBAC	App guardrails	Proof-of-Control
Enforcement	Probabilistic alignment	Binary API gating	Application code	Hardware-isolated policy evaluation
Context awareness	Context-window scope	None (context-blind)	Partial (local state)	Complete (snapshot + path)
Tamper resistance	Negligible (injection-prone)	Moderate (credential-reliant)	Low (bypassable)	Cryptographic (TEE-backed)
Audit verification	Unauthenticated text	Operator-mutable logs	Operator-mutable logs	Machine-checkable Merkle proofs
Operator trust	Total	Total	Total	Zero-trust (hardware-anchored)

Table 2: Governance mechanisms compared. Only Proof-of-Control clears the binary threshold of Section 3.2.

5 Formal Model

Let $\mathcal{A}$ be the space of agent actions, $\mathcal{I}$ the space of cryptographically verifiable agent identities, and $\mathcal{S}$ the space of host state snapshots. The executed steps up to time t form the partial path $p_t = \langle a_0, a_1, \dots, a_{t-1} \rangle \in \mathcal{A}^*$. A compliance policy is a deterministic evaluation function

$$\Pi : \mathcal{A} \times \mathcal{I} \times \mathcal{S} \times \mathcal{A}^* \longrightarrow \mathcal{D}, \quad \mathcal{D} = \{\text{ALLOW}, \text{DENY}, \text{MODIFY}(a'), \text{ESCALATE}(h)\}, \quad (2)$$

where $a' \in \mathcal{A}$ is a sanitized payload and h a human-oversight routing profile. Proof-of-Control requires Π to execute inside an isolated enclave, producing per-step proof

$$\pi_t = \text{Sign}_{k_{\text{AE}}}(H(\sigma_t^{\text{snap}}) \parallel d_t \parallel t \parallel L_t), \quad (3)$$

with k_{AE} accessible exclusively to the Attesting Environment. The path argument makes the policy *path-aware*: Π evaluates the proposed action against context carried from upstream invocations, so composition attacks that are invisible to per-action evaluation are within the decision boundary [6]. A deliberate *determinism boundary* limits claims: Proof-of-Control verifies deterministic facts of execution (which model ran, on what input, under which authority, producing which effect); it does not represent output correctness, model reasoning, fairness, or future behavior as verified. Verification is performed, not validation [36].

6 Threat Model

Proof-of-Control assumes a zero-trust model in which the host OS, network, and operator may be compromised. Representative threats and mitigations:

- **Compromised host operator** (suppressing enforcement, editing logs, issuing unauthorized calls): policy logic and signing keys are enclave-isolated; altering the policy manifest changes the enclave measurement, causing verifiers to reject evidence [20].
- **Prompt injection**: hooks intercept proposed calls before dispatch; the enclave evaluates deterministic rules independently of LLM reasoning, denying unpermitted actions inline [5].
- **Log alteration and omission**: Merkle chaining makes deletion or reordering break the signed root; monotonic sequencing makes silent omission detectable [10].
- **Cascading sub-agent exploitation**: delegation hooks enforce constraint inheritance; sub-agents must produce evidence verified through multi-verifier topologies before actions execute [35].
- **Skill-composition risk**: path-aware authorization (C4) gates capability flow and trust transfer that artifact-level vetting cannot see [6].
- **Monitoring blind spots**: trajectory validators must be evaluated for structured-trace parsing competence, not only natural-language safety performance [7].

The full normative threat model enumerates 32 threats drawn from MITRE ATLAS [37], NIST AI 100-2, and the OWASP catalogues, grading per-threat coverage and stating explicit out-of-scope boundaries; four rows are deliberately marked *not addressed* (e.g., output correctness), which is what keeps the claim honest.

7 Regulatory Coverage: A Reproducible Mapping

Following the coding methodology of HAARF [11], every PoC requirement was coded against eight frameworks as Exact Match (equivalent clause in scope and intent), Partial Match (same topic without operator-independent evidence, or at lesser depth), or No Match. Coverage = (EM + PM)/111. Coding is single-coder seed data at section granularity, published with per-row rationales and clause attributions for working-group validation; the computation is reproducible from the public coding sheet.

Framework	Exact	Partial	None	Coverage
NIST AI RMF [38]	0	71	40	64%
ISO/IEC 42001 [36]	8	60	43	61%
OWASP AISVS [39]	16	51	44	60%
EU AI Act [28]	8	55	48	57%
SOC 2 [40]	8	55	48	57%
CSA AARM [41]	11	44	56	50%
Zero Trust (SP 800-207) [18]	8	41	62	44%
MITRE ATLAS [37]	0	30	81	27%

Table 3: Coverage of the 111 Proof-of-Control requirements by external frameworks (draft seed coding). The exact-match band is thin everywhere: existing frameworks require the *controls* but almost never operator-independent *evidence* that they held—the gap between Tier 2 and Tier 3.

Two patterns are consistent. First, the partial-match band is wide and the exact-match band thin: NIST AI RMF attains the highest coverage with zero exact matches, because governance frameworks ask for nearly every control PoC verifies while never requiring mechanism-generated evidence. Second, the no-match gap concentrates in the evidence-property, tier-grading, and disclosure chapters: no coded framework grades evidence by how independently it can be verified, requires trust-assumption disclosure, or reaches self-enforcing execution. The gap is, by design, the standard’s reason to exist.

8 Standards Integration Roadmap

ISO/IEC JTC 1/SC 42. Proof-of-Control provides a technical realization for active SC 42 workstreams [12, 13, 42]: a technical annex for auditable runtime controls under ISO/IEC 42001 Clauses 8–9 (supplying cryptographically verifiable compliance evidence during certification audits) [36], operationalization of ISO/IEC 23894 risk treatment as executable, hardware-verified policy bundles [43], and an SC 42/SC 27 bridge coupling AI oversight with confidential-computing standards [44].

IETF RATS. An Internet-Draft establishing an AI-agent EAT profile [9, 34]—standardizing claims for lifecycle hook types, policy-bundle hashes, Agent Bill of Materials (AgBOM) digests, and step indices—and an extension of multi-verifier attestation to sub-agent delegation chains [35].

Agent Control Specification. Proof-of-Control supplies the missing cryptographic layer for ACS [15, 16]: anchoring its decision engine in TEEs and signing decision tokens.

NIST AI 600-1. Proof-of-Control provides concrete implementations for the generative-AI profile’s measurement controls [17, 45, 46]—enforcing that deployment configurations cannot bypass security controls (hardware-isolated policy boundaries) and satisfying continuous security verification (attestation tokens for every tool call and state transition).

9 Performance Budget

Because agent workflows involve synchronous tool invocations, governance must operate at machine speed. Proof-of-Control targets end-to-end overhead below **15 ms per intercepted step**, achieved

by (i) policy engines compiled to native WebAssembly/Rust modules executing in enclave memory, eliminating IPC overhead; (ii) hardware-accelerated Ed25519/ECDSA P-256 signatures; and (iii) asynchronous anchoring—leaf generation and signing are inline, while publishing Merkle roots to external ledgers runs in the background. The verification-bandwidth economics motivating this budget are structural: as execution cost falls toward zero, verification becomes the scarce factor [4], so checking compliance must be cheap, binary, and mechanical.

10 Implementation Roadmap

Phase 1 (months 1–6): open specification and reference core—core schema extensions on the ACS repository, open-source Rust reference enclave policy engines, and the IETF Internet-Draft for the AI-agent EAT profile. **Phase 2 (months 7–12): multi-vendor interoperability and SDO balloting**—SC 42 working groups, conformance testing across Python, Node.js, and Rust agent frameworks, and telemetry integration with OpenTelemetry and OCSF. **Phase 3 (months 13–24): ratification and deployment**—normative annexes in ISO/IEC 42001 and 23894, RFC publication, and deployments across financial, healthcare, and public-sector workloads. The standard’s adopter-side rollout mirrors this in three phases from foundational key/signing/logging infrastructure through expanded proof coverage to continuously monitored operation.

11 Limitations and Open Questions

Four limitations are stated deliberately. (1) The determinism boundary: PoC verifies execution facts, not output correctness, fairness, or intent; validation remains a human responsibility. (2) The binary threshold is under dedicated cryptographic and complexity-theoretic review, including impossibility results on what agentic verification can achieve efficiently. (3) The regulatory mapping is a single-coder seed pending multi-coder validation and inter-coder agreement statistics. (4) Tier-4 self-enforcement creates an availability dependency (the system halts when proofs cannot be produced) that deployments must assess and disclose. Open working-group questions—identity/authorization ownership, verifiable-but-unlinkable identity, evidence continuity across jurisdictions as a possible fifth evidence property, and continuous-monitoring cadence—are tracked publicly in the specification’s open-issues register.

12 Conclusion

The claims-based world, in which relying parties accept an operator’s word for what an autonomous system did, does not scale to agents no one can watch. Proof-of-Control replaces attestation with mechanism-generated, hardware-anchored, independently checkable evidence: six domains of verifiable facts, a binary threshold that makes verifiability procurable, an architecture that intercepts every consequential action, and a coverage mapping demonstrating that no existing framework occupies this layer. The specification, audit checklist, coding sheets, and reference tooling are developed openly under the Advanced AI Society and are freely licensed for implementation.

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